Database Management - Exercises

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Exercise: Data Management

Learning Objectives

1. A database is an organized and structured set of records stored and managed for a common purpose.
2. Big data primarily refers to data sets that are too large or complex to be dealt with by traditional data-processing software.
3. Enable multi-user operations: Manage transactions, query and manipulate data sets, ensure data consistency and integrity automatically and retrieve large data quantities.

Introductory Example: Movie Recommendation Dashboard

2.

Edit the SQL code behind the Movie Recommendation question so that only 10 Movies are recommended.

```sql
SELECT MovieID, Title, COUNT(*) AS N5R
FROM Ratings
WHERE UserID IN (
SELECT UserId AS TargetGroup
FROM Ratings
WHERE MovieID = ( SELECT MovieID
        FROM Movies
        WHERE {{filter}} LIMIT 1 )
 )
GROUP BY MovieID
```

```sql
ORDER BY COUNT(*) DESC
LIMIT 10; -- Change value here!
```

3.

Edit the SQL code behind the Genres question so that the genres are sorted by their corresponding number of movies from top to bottom.

```sql
SELECT Genres.Name, COUNT(*) as n_movies -- Set var name
FROM Movies
JOIN Movie_has_Genre USING (MovieID)
JOIN Genres USING (GenreID)
GROUP BY Genres.Name
ORDER BY
    n_movies DESC; -- Order by n_movies
```

4.

Add a new chart to the dashboard that displays the development of movies per genre and per decade (10 year bins) between the 1950ies and the 2010s.

```sql
SELECT
    Genres.Name AS Genre,
    CAST(strftime('%Y', Release_date) / 10 AS INT) * 10 AS Decade,
    COUNT(*) AS n_movies
FROM
    Movies
JOIN
    Movie_has_Genre USING (MovieID)
JOIN
    Genres USING (GenreID)
WHERE
    strftime('%Y', Release_date) BETWEEN '1950' AND '2019'
GROUP BY
    Genre,
    Decade
ORDER BY
    Decade ASC,
    n_movies DESC;
```

> To achieve the correct visualisation, you need to edit the chat in Metabase itself.